Practical Considerations

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The starting point

Start from a clear question

First things first

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- What is the best way to extract my samples?

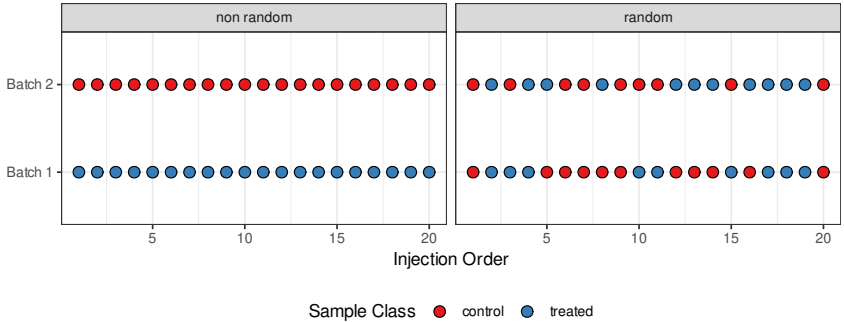
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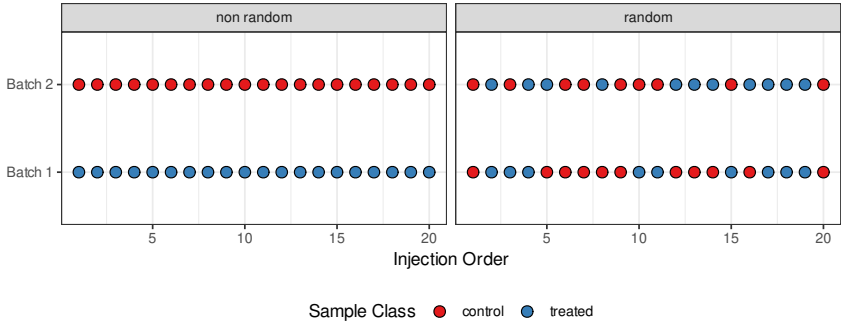
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- **MS performance degrades with time**: how long can I run my sample list before cleaning

Controlling analytical variability



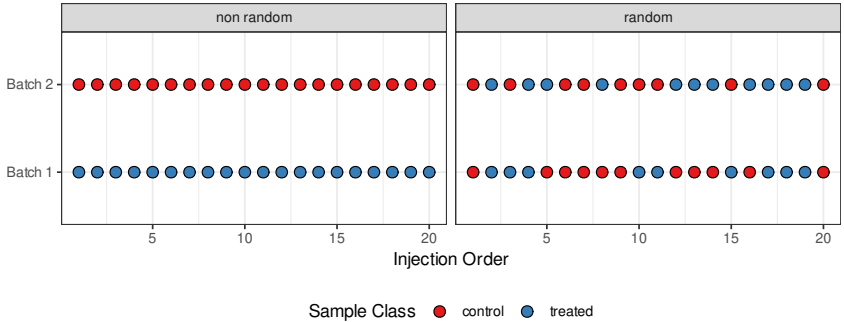
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- **Measure the analytical variability**

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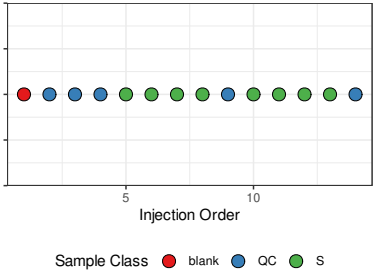
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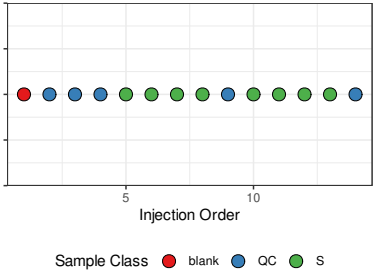
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- QC samples should be representative of the chemical complexity of the samples
- **Pooled samples** are often the preferred choice

A fairly good sample list



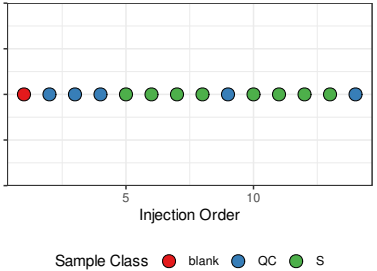
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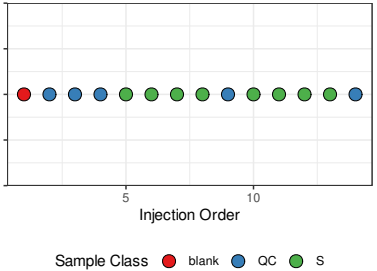
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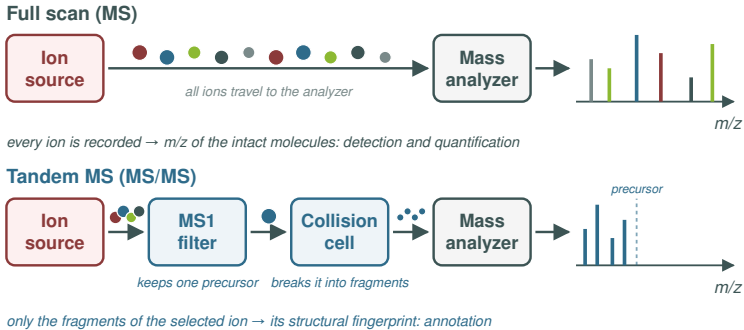


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- One could add QC dilutions
- Also the design of the sequence can be validated

Other relevant points of discussion

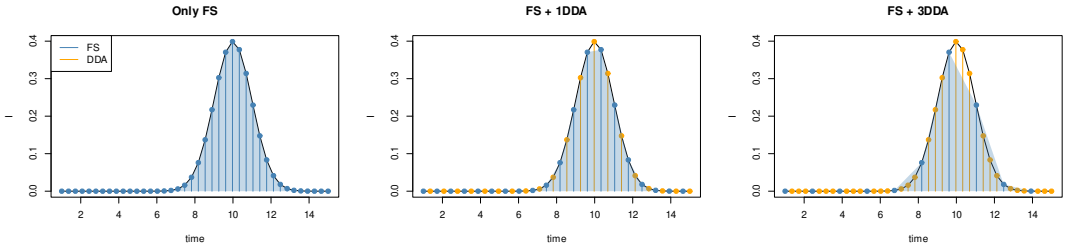
- Standards and blanks
- Analytical batches and batch effect
- Randomization, analytical batches and large studies in clinical settings

DDA and Fragmentation



Experiments combining full scan and fragmentation are quite popular, because fragmentation patterns are needed for metabolite annotation and for setting up quantitative methods

Use your time wisely



The more experiments I do, the worse my estimate of the peak area ...

- plan different runs for FS and DDA
- fragment only ions of interest
- strike a balance between chromatography, instrument speed and number of experiments